

Innovation-Safe Compression of Financial Strategy Libraries: Semantics, Complexity, and Algorithms

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Abstract

A portfolio research team can reduce maintenance burden by retiring strategies, but an entry that adds nothing to today's trading menu may still enable valuable future research. We ask which entries can be removed while preserving both uses. We give a conditional preservation principle: an operating frontier and a declared closure of reusable capabilities preserve the controlled research and operating process under explicit invariance assumptions. With identity closure, minimum-burden safe retention is exactly classical weighted set cover. This representation supplies preprocessing rules, exact methods, and a transferred greedy guarantee. Counterexamples show that preserving current operation can destroy future research value and that heaviest-safe-first deletion can be arbitrarily far from minimum burden. The computational benchmark retains all 3,907 run outcomes and verifies exact-method agreement wherever the required comparisons are available. A retrospective financial study finds large burden-index reductions and additional research options, but no policy replacement in common equities and a loss from its sole ETF replacement. A calibrated synthetic experiment distinguishes option availability from reliable exercise: a small positive bridge value has a learned-value interval spanning zero, while a larger designed dose has a positive interval. These results establish a specified retention principle and bounded experimental findings, without establishing a market premium.

Keywords: financial strategy libraries, innovation-safe compression, weighted set cover, exact algorithms, model governance

1 The financial retention problem

A portfolio research team accumulates strategies that it must test, monitor, document, and maintain. Some are used in portfolios today. Others remain in the library because their signals, filters, allocation rules, or risk controls may be useful in later research. Maintaining every strategy consumes resources; removing a strategy can reduce that burden. The manager’s problem is to decide which strategies can leave the active library without reducing its best modeled operating value or losing declared capabilities needed to build future strategies.

Consider two strategies that offer the same current portfolio opportunity. One also carries a capability needed by a possible new strategy. If the team retains only the other strategy, today’s opportunity is preserved but the new strategy may become unavailable. A performance-based redundancy check cannot detect that loss. Conversely, keeping every strategy that carries a useful capability can be unnecessarily expensive when several entries preserve the same capabilities. The decision therefore concerns the retained library as a whole. Figure 1 gives a concrete example.

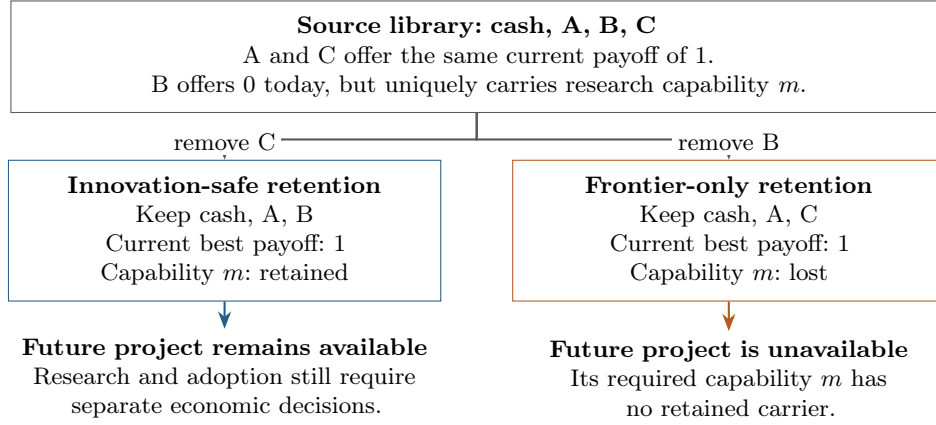


Fig. 1 Current operating redundancy can conceal a research capability. This exact schematic has one operating condition, identity closure, unit burden for A, B, C, and zero burden and payoff for mandatory cash. Both retained libraries have burden 2 rather than the source’s 3 and preserve its frontier. The frontier-only choice is augmented to the same burden; it is not asserted to be the minimum frontier-only library. Only B carries *m*, which is required for the future project. Availability is preserved without assuming that the project is profitable or will be selected. These are model values, not market observations.

Here, removal means leaving the active maintenance and validation set. Legal records, source code, research history, and mandatory archives remain subject to their own retention rules. A capability is an explicitly declared input to the modeled research process. The model applies when retaining its designated carriers is what makes that capability available; it does not assume that a strategy label contains all of an institution’s knowledge.

We represent the manager’s two preservation targets in simple terms. The *operating frontier* records the best current operating value offered by the source library

in each modeled condition. The *generative closure* records the capabilities available after applying the declared rules for combining the retained strategies’ components. Innovation-safe compression finds a least-burden subset of the source library that preserves both targets. The burden can represent a specified validation or governance index; its interpretation must be fixed before looking at subsequent investment outcomes.

The contribution is a conditional preservation principle for a maintained strategy library, an exact optimization representation of that principle, and counterexamples showing why weaker deletion rules can fail. The financial question concerns the availability of reusable inputs to subsequent research. The covering algorithms used to implement the retention decision are established methods. Our financial evidence measures the declared burden and research opportunities separately from subsequent investment performance.

The mathematical argument follows that decision. First, we specify when the two targets summarize the library’s role in a research process. We prove that preserving them then preserves the process’s operating rewards and future choices. An identity-dependent project supplies a counterexample when the assumption fails. Next, for a model in which capabilities are simply the union of retained components, we show that the decision is a weighted covering problem. This connects the financial specification to established optimization methods (Karp 1972; Chvátal 1979). A separate counterexample proves that repeatedly deleting the heaviest redundant entry can leave an arbitrarily expensive library even when every deletion is safe.

The computational study checks exact agreement and the burdens returned by several methods on a registered synthetic design. The financial application then examines portfolio-strategy libraries formed from point-in-time common equities and ETFs. It finds substantial reductions in a validation-burden index while preserving more research options than a budget-matched library that preserves only current operation. The screening policy exercises none of the additional common-equity options; its single ETF replacement loses held-out value. A calibrated synthetic experiment separately shows why an available positive-value option can remain too difficult to recognize at a small signal-to-noise ratio.

These results distinguish preserving an opportunity, deciding to use it, and realizing value from it. They establish a conditional retention principle and bounded computational and financial findings. The financial application is retrospective development evidence, and its burden index is not a measured institutional cost saving. The assumptions and the counterexamples below state precisely where the principle applies.

2 A model of operation and future research

The model records the strategies a manager can operate and the capabilities available for further research. The controller observes a filtered belief, operates from a verified strategy library, and may initiate positive-duration research that generates and verifies a candidate before admission. The theorem does not derive library invariance from finance primitives; it identifies the conclusion that follows when the primitives pass the explicit factorization checks below. The retention decision is made once, before the controlled process is evaluated.

2.1 Hidden state, filtered belief, and operating profiles

Let X be a finite hidden market-state set with at least two elements and let B be a finite nonempty set of filtered beliefs. A belief $b \in B$ has an exact rational interpretation $\mu_b \in \Delta_{\mathbb{Q}}(X)$; the already-filtered belief evolves under an exact rational Markov kernel P_B . The controller observes b , not the hidden state $x \in X$.

The finite strategy catalog is S , with distinguished inactive strategy s_0 . Strategy s has immutable state-contingent operating payoff $u_s : X \rightarrow \mathbb{Q}$ and belief-indexed operating profile

$$j_s(b) := \sum_{x \in X} \mu_b(x) u_s(x), \quad j_{s_0}(b) = 0. \quad (2.1)$$

Thus $j_s(b)$ is the payoff from operating s now; research costs and continuation values are separate objects. The formal projection theorem uses the resulting exact profile table as primitive data, so it does not claim a kernel-verified derivation of that table from a hidden-state model.

Partial information matters to the controlled projection through the filtered belief and its transition law. It is not required by the later tagged-cover equivalence, which needs only a finite index set of operating-profile rows. The financial study in Section 8 uses calendar-year performance profiles crossed with declared cost and risk settings as these rows; it does not estimate the hidden-state filter in this subsection.

2.2 Modules, closure, and the raw retained library

Let M be a finite module set. Each strategy carries an immutable set $\text{mods}(s) \subseteq M$, with $\text{mods}(s_0) = \emptyset$. A module is a declared reusable capability tag for later research, not a direct operating payoff or necessarily a separately owned software artifact. The declared closure operator

$$\text{cl} : 2^M \longrightarrow 2^M$$

is extensive, monotone, and idempotent. These properties allow capabilities to be generated from combinations of raw modules; identity closure is the special case $\text{cl}(A) = A$.

A raw retained library is a source subset $L \subseteq S$ containing s_0 . Its operating frontier, raw module inventory, and generative closure are

$$F_L(b) := \max_{s \in L} j_s(b), \quad U_L^{\text{raw}} := \bigcup_{s \in L} \text{mods}(s), \quad (2.2)$$

$$C_L := \text{cl}(U_L^{\text{raw}}). \quad (2.3)$$

Finiteness and mandatory inactivity make the maximum attained and imply $F_L(b) \geq 0$. The raw decision state is (b, L) : continuing earns the frontier payoff, advances the belief once, and leaves the library unchanged.

An entry may itself be a portfolio strategy, but the operating decision here selects one catalog entry at a time. Equality of F_L therefore preserves the best entry's modeled payoff at each belief; it does not by itself preserve the risk–return opportunities from combining entries with different joint returns. An application that reallocates

across entries must include the relevant joint payoff and allocation information in its preservation target.

2.3 Generation, verification, admission, and timing

Let Q be a finite project set. At project q , belief b , and closure C , candidate generation follows an exact law

$$G(q, b, C) \in \Delta_{\mathbb{Q}}(\{\text{none}\} \dot{\cup} S).$$

A generated candidate s passes verification with exact probability $\nu(q, b, C, s)$. The admitted law is derived rather than separately chosen:

$$\Gamma(q, b, C)(s) = G(q, b, C)(s)\nu(q, b, C, s), \quad (2.4)$$

and all failed-generation and failed-verification mass is assigned to **none**. Successful admission adds the verified catalog row, whereas every other outcome leaves the source unchanged:

$$L \oplus \text{none} = L, \quad L \oplus s = L \cup \{s\}. \quad (2.5)$$

Each available project has an initiation cost, an operation/suspension flag, and an integer duration $d_q \geq 1$. Its exact joint completion law couples the intervening belief path with the admitted outcome; the two components need not be independent. The incumbent library remains in force until completion, then (2.5) is applied. Full probability-table definitions, null-outcome normalization, path and outcome marginals, and pushforward identities are assigned to Online Resource 1, Section S1.

2.4 The productive library state

Definition 2.1 (Frontier–closure library state). The productive state of a retained library is

$$K_L := (F_L, C_L). \quad (2.6)$$

It is the *library component* of the controlled state at embedded decision epochs. The filtered belief remains part of that state. No new decision is taken during a committed project. If a calendar-time representation exposes intermediate observations, it must also retain enough history to condition the declared joint completion law; project identity and remaining duration alone need not suffice for an arbitrary path–outcome coupling.

For $K = (F, C)$, define the local compressed update

$$\text{addK}(K, \text{none}) = K, \quad \text{addK}(K, s) = (b \mapsto \max\{F(b), j_s(b)\}, \text{cl}(C \cup \text{mods}(s))).$$

The three defining closure properties give

$$K_{L \oplus o} = \text{addK}(K_L, o). \quad (2.7)$$

Assumption 2.2 (Finite library-invariant research process). The sets B, S, M, Q are finite, B and every admissible library are nonempty, and all displayed profiles and

probability tables are exact. The inactive strategy has zero profile and no modules; closure is extensive, monotone, and idempotent. Generation and primitive verification depend on the incumbent library only through (q, b, C_L) . Project availability, initiation cost, duration, operation flag, and the joint completion law depend on the library only through (b, K_L) . Each duration is positive, the joint law has the declared belief-path and admitted-outcome marginals, and discount satisfies $0 \leq \beta < 1$. Stationary conclusions additionally assume the recorded raw and compressed Bellman operators are contractions.

3 What safe retention preserves

Theorem 3.1 (Raw-to-compressed controlled Markov projection). *Under Assumption 2.2, the update in (2.7) induces a normalized transition on realizable states K_L . If $K_L = K_{\bar{L}}$, the two raw libraries have identical project availability, cost, duration, incumbent operating reward blocks, and joint terminal-belief-next-compressed-state laws after projection. Hence (b, K_L) is a controlled semi-Markov projection at embedded decision epochs and, for every finite calendar horizon h ,*

$$V_h^{\text{raw}}(b, L) = \bar{V}_h(b, K_L).$$

Under the additional contraction clause, stationary fixed-point values agree under the same projection and a maximizing stationary compressed selector lifts to a Bellman-optimal raw selector.

Proof sketch. Equation (2.7) makes the next compressed state a deterministic pushforward of the admitted raw update. The library-invariance assumptions make this law, every feasible action, and its payoff block constant on each fiber of $L \mapsto K_L$. Strong induction on calendar horizon is well founded because every research duration is positive; corresponding raw and compressed action values, and then their finite maxima, agree. Bellman intertwining and contraction give the stationary clauses. The complete normalization and projection proof is in Online Resource 1, Sections S1–S2. The result permits belief-admission dependence and makes no generic quotient-minimality claim. Action correspondence concerns Continue and research-project choices. It does not require retaining every raw strategy identity that attains the same operating frontier.

Operational factorization audit.

The assumption can be checked table by table. Project availability, cost, duration, operation while active, and the joint completion law must be constant whenever (b, F_L, C_L) is constant; candidate generation and verification must be constant whenever (q, b, C_L) is constant; and admission must update the compressed state through addK. A nontrivial finite class satisfying these conditions consists of module-gated projects whose availability is a predicate of (q, b, C_L) , whose candidate and verification tables are indexed by (q, b, C_L) , and whose duration, cost, and completion tables are indexed by (q, b, F_L, C_L) . Correlation between terminal belief and admission remains unrestricted within each indexed joint table.

The condition is substantive. Let two one-active-strategy libraries $L_a = \{s_0, a\}$ and $L_b = \{s_0, b\}$ have identical zero profiles and no modules. Then $K_{L_a} = K_{L_b}$. If a zero-cost, one-period project with deterministic positive completion value is available only when the raw identity a is retained, their feasible actions and values differ. The projection therefore fails. The same failure occurs if a primitive depends directly on raw cardinality, provenance, owner, or archival status. These variables must either be included in the compressed state or ruled out by the declared information architecture.

3.1 Retention burden and exact source-relative compression

Let $w_{s_0} = 0$ and give every active strategy an exact rational retention weight $w_s > 0$. The one-time burden is

$$W(L) := \sum_{s \in L} w_s. \quad (3.1)$$

It may index storage, validation, monitoring, active maintenance, retrieval, governance, and human attention. Additivity is a modeling assumption. A weight is not automatically an observed monetary or labor cost, and it is not operating payoff or project cost. Burden is not a component of K_L : equal productive states can have different burdens. In the financial cases, “retained” means retained in the active maintenance and validation set; mandatory archival records, code, data, ownership, and reactivation controls remain outside this binary decision.

For a fixed verified source L , exact source-relative compression chooses an inactive-containing sublibrary $L' \subseteq L$ with $K_{L'} = K_L$ and minimum $W(L')$. By Theorem 3.1, this outer retention decision changes only the raw representation, not the declared productive process. Section 4 defines its feasible family and optimization model.

4 Exact compression, covering representation, and complexity

Fix a verified finite source library L . Safety is source-relative: compression may delete source entries but may neither import a replacement nor change the declared profiles, modules, or closure operator.

4.1 Safe feasibility and productive preservation

The safe-feasible family and its global minimum-weight correspondence are

$$\mathcal{P}_{\text{safe}}(L) := \{L' \subseteq L : s_0 \in L', F_{L'} = F_L, C_{L'} = C_L\}, \quad (4.1)$$

$$\mathcal{M}_{\text{safe}}(L) := \arg \min_{L' \in \mathcal{P}_{\text{safe}}(L)} W(L'). \quad (4.2)$$

Equality is exact at every declared belief and for the entire declared generative closure. In particular, a currently dominated strategy is not safe to remove merely because it contributes no current frontier value.

Theorem 4.1 (Minimum-resource innovation-safe representation). *Let L be a finite source library containing s_0 , let the catalog and closure satisfy Assumption 2.2, and let $w_{s_0} = 0$ with exact rational $w_s > 0$ for every active source strategy. Then $\mathcal{P}_{\text{safe}}(L)$ is finite and nonempty, $\mathcal{M}_{\text{safe}}(L) \neq \emptyset$, and every $L' \in \mathcal{P}_{\text{safe}}(L)$ has the same finite-horizon productive values as L . Under the contraction clause of Assumption 2.2, it also has the same stationary productive value and complete optimal stationary-action correspondence.*

Proof. The source is feasible relative to itself and has only finitely many sublibraries, so an exact burden minimum is attained. Feasibility is exactly $K_{L'} = K_L$; the value and action conclusions follow from Theorem 3.1. Neither optimizer uniqueness nor a particular deletion path follows.

A converse from equal observed behavior to equal closure requires an additional detectability assumption; the supporting proof states that separate condition.

4.2 Identity closure induces a tagged cover

The reduction in this subsection is the computational consequence of the preservation semantics. Once the tagged universe is constructed, the resulting optimization problem is ordinary weighted set cover: its generic binary model, NP-hardness antecedent, greedy approximation theorem, mask recurrence, and standard cover preprocessing are not claimed as new combinatorial-optimization theory. The application-specific content is the exact mapping from frontier and closure equality to tagged coverage, together with its identity-closure limit.

The algorithms below apply to identity closure. General closure remains governed by exact equality of the closed module unions.

Assume throughout this subsection that closure is identity. Define the disjoint tagged requirement universe

$$\mathcal{U}_L^{\text{tag}} := \{\text{frontier}(b) : b \in B\} \dot{\cup} \{\text{module}(m) : m \in C_L\}, \quad (4.3)$$

and, for each $s \in L$, its source-relative coverage

$$\mathcal{R}_s := \{\text{frontier}(b) : j_s(b) = F_L(b)\} \cup \{\text{module}(m) : m \in \text{mods}(s)\}. \quad (4.4)$$

Frontier and module tags remain distinct even when their underlying identifiers print alike. Exact equality in the first term means that every tied source attainer covers the belief row.

Theorem 4.2 (Tagged-cover equivalence under identity closure). *Let B, M, S be finite, let $B \neq \emptyset$, and let $s_0 \in L' \subseteq L \subseteq S$. Profiles are exact rational functions $j_s : B \rightarrow \mathbb{Q}$; $j_{s_0} = 0$, $\text{mods}(s_0) = \emptyset$, maxima are attained, and closure is identity on 2^M . Then*

$$F_{L'} = F_L \text{ and } C_{L'} = C_L \iff \mathcal{U}_L^{\text{tag}} \subseteq \bigcup_{s \in L'} \mathcal{R}_s. \quad (4.5)$$

No unique frontier attainer or unique module carrier is assumed. If $F_L(b) = 0$, mandatory s_0 itself covers $\text{frontier}(b)$.

Proof. If L' preserves the frontier, a finite maximizer in L' covers each frontier tag. Under identity closure, $C_{L'} = C_L$ is equality of raw module unions, so some retained carrier covers every module tag. Conversely, source inclusion gives $F_{L'} \leq F_L$ and $C_{L'} \subseteq C_L$. Coverage supplies a retained source attainer at every belief and a retained raw carrier for every source module, giving the reverse inclusions. Ties and multiple carriers are existential choices, while the zero-frontier row is witnessed by s_0 .

For exact rational weights, the equivalent binary model is

$$\min_x \sum_{s \in L} w_s x_s, \quad (4.6)$$

$$\text{s.t.} \quad \sum_{s \in L: j_s(b)=F_L(b)} x_s \geq 1, \quad b \in B, \quad (4.7)$$

$$\sum_{s \in L: m \in \text{mods}(s)} x_s \geq 1, \quad m \in C_L, \quad (4.8)$$

$$x_{s_0} = 1, \quad x_s \in \{0, 1\}, \quad s \in L. \quad (4.9)$$

The source restriction is built into the columns. All zero-frontier rows may be removed after fixing s_0 , but retaining them in $\mathcal{U}_L^{\text{tag}}$ keeps the semantic construction literal.

The boundary is substantive. If $\text{cl}(\{a, b\}) = \{a, b, c\}$ although no strategy carries c , an independent row for each member of C_L rejects even the source library. General closure therefore requires direct closure equality or a separately proved extended formulation. Frontier rows alone and module rows alone are likewise insufficient; exact counterexamples are in Online Resource 1, Section S3.

4.3 Baseline mixed-integer formulation

Equations (4.6)–(4.9) are the baseline MIP. Its input is a validated, canonically ordered identity-closure instance with exact weights and declared mandatory columns. Exact preprocessing may fix columns and remove satisfied rows before JuMP and HiGHS receive the residual binary cover. Exact weights are scaled to an integer objective only after the scaling identity and solver range are verified.

Each returned selection is lifted to original strategy identities and independently checked for mandatory retention, frontier equality, closure equality, and exact burden. Fractional vectors are rejected without rounding. Solver status and bounds remain numerical evidence of global optimality; the exact postcheck certifies the returned library's feasibility and burden. If preprocessing solves the instance completely, no solver is invoked.

4.4 Computational complexity

IDENTITY-CLOSURE SAFE COMPRESSION (ISC-ID) has the following explicit decision encoding. The input lists a finite source $L = \{s_0, s_1, \dots, s_n\}$, a finite nonempty

belief set, exact binary-rational profiles, a finite module universe and incidence rows, identity closure, positive binary-integer active weights, and a nonnegative binary-integer threshold D . Strategy s_0 is mandatory, has zero profile, no modules, and zero weight. The question is whether some $s_0 \in L' \subseteq L$ preserves the exact source frontier and closure and has $W(L') \leq D$.

Theorem 4.3 (Identity-closure safe-compression complexity). *Under the explicit encoding in the preceding paragraph, ISC-ID is NP-complete. It remains NP-complete on the restricted subclass with one belief, zero profiles for every strategy, a zero-weight module-free inactive strategy, identity closure, a source whose active module rows have full union, and unit active weights. Consequently, minimum-burden identity-closure safe compression is NP-hard, and any broader model class containing this subclass is NP-hard. No NP-membership conclusion for arbitrary nonidentity closure follows unless its encoding admits a polynomial exact closure-equality verifier.*

Proof sketch. A selected-strategy bit vector is a polynomial certificate: exact rational maxima, row unions, binary-integer burden, source inclusion, and mandatory retention are polynomially checkable in the explicit input length. For hardness, normalize Karp’s SET COVER decision problem (Karp 1972) to a nonempty universe and a full-union indexed family, using fixed yes/no outputs for the empty- or uncovered-universe branches. Give each universe element a module, each indexed set an active strategy carrying its modules, every active strategy unit weight and zero profile, and add one zero-profile inactive strategy. Use one belief and copy the budget. The inactive strategy preserves the trivial frontier, while identity-closure equality is exactly set-cover equality and burden is exactly selected-set cardinality. Incidence size and integer bit lengths are copied polynomially. The complete membership, normalization, forward, reverse, size, and threshold audit is in Online Resource 1, Section S3. No strong-hardness, inapproximability, or APX-hardness claim is made.

4.5 Exact preprocessing of the tagged cover

Fix the tagged universe from the source before reducing the model. After declared mandatory strategies are selected, let Q be the unsatisfied rows, A the available nonfixed columns, and $\mathcal{R}_s^Q = \mathcal{R}_s \cap Q$. Mandatory weights are nonnegative, optional weights are strictly positive exact rationals, and there are no identity-sensitive side constraints, groups, exact-cardinality conditions, or mutual exclusions beyond the pure cover. All incidence and weight comparisons are exact.

Proposition 4.4 (Exact preprocessing suite). *Under the assumptions in the preceding paragraph, the following fixed-point rules preserve feasibility, the exact optimum value including forced-weight offsets, and at least one liftable optimum:*

1. fix the unique available carrier of any residual requirement;
2. delete nonmandatory s when $\mathcal{R}_s^Q \subset \mathcal{R}_t^Q$ and $w_t \leq w_s$;
3. for identical residual coverage, retain a stable minimum-weight representative and record every equal-weight substitution;

4. after a forced selection, remove satisfied rows and recompute all rules;
5. delete a nonmandatory column with empty residual coverage; and
6. merge rows having identical carrier sets.

Strictly cheaper dominance, strictly heavier duplicates, empty positive-weight columns, forced columns, and duplicate rows preserve all original optimizer identities. Equal-weight duplicate columns recover all identities through the recorded substitution map. Equal-weight strict dominance need not preserve all optimizer identities and makes no such claim. Restarting the stable scan after each transformation terminates after finitely many row or column removals and yields a fixed point. A zero-carrier residual row is an infeasibility certificate, not an invitation to repair the instance.

Proof sketch. A unique carrier is forced by its covering row. A no-heavier coverage superset can replace a dominated selected column without losing a witness. Identical columns permit exact substitution; identical rows impose the same inequality; and an empty positive-weight column occurs in no optimum. Forced columns contribute a selection-independent offset, and propagation composes these correspondences. Every successful step removes a finite row or column, which proves termination. The strict/equal weight cases give precisely the optimizer-identity qualifications in the statement. Full propositions, adversarial examples, reconstruction maps, and the deterministic rule order are in Online Resource 1, Section S4.

5 Algorithms and guarantees

All methods in this section consume the same validated identity-closure instance. Strategies and tagged requirements have canonical original order; weights are exact rationals; mandatory flags, source profiles, raw module memberships, the source frontier, and the source closure are stored independently of the incidence matrix. Preprocessing supplies an exact objective offset and a reconstruction map. The number of residual optional strategies is n , the number of residual tagged requirements is r , and strategy i has residual mask R_i and positive weight w_i .

5.1 Common output and certificate

Every algorithm returns original strategy identifiers in canonical order, the exact burden, its declared completion status, its tie rule, an algorithm trace or search counters, and any required random seed. A result is reportable only after an independent certificate recomputes from the original catalog:

1. every mandatory strategy is retained;
2. every tagged requirement is covered;
3. the complete source operating frontier is preserved exactly;
4. the complete source identity closure is preserved exactly; and
5. the exact burden agrees with the returned burden.

Passing this certificate establishes feasibility and burden for the returned library. It does not establish heuristic optimality, convert a solver status into a proof, or show that

two implementations are independent. Complete-tie output is permitted only when the preprocessing map declares every optimizer identity reconstructable; otherwise the result is one explicitly declared representative.

Table 1 Guarantees under the stated assumptions. Exact returned-library checks certify feasibility; they do not turn solver or heuristic output into a proof of optimality.

Method	Guarantee and limit
Enumeration	Exact optimum; exponential in optional strategies n .
Mask dynamic programming	Exact optimum; $O(n2^r)$ recurrence operations, exponential in residual requirements r .
HiGHS mixed-integer model	Exactly rechecked feasible incumbent; solver-reported optimality is separate evidence.
Weighted greedy, with optional reverse deletion	$H(d) \leq H(r)$ approximation under identity closure and positive optional weights.
Cardinality greedy	Feasible cover; no burden factor for unequal weights.
Rechecked deletion, including multi-start	Feasible, one-deletion irreducible endpoint; no general burden factor. Heaviest-first has an unbounded worst-case ratio.

5.2 Independent exact methods

5.2.1 Complete enumeration

Proposition 5.1 (Correctness of complete enumeration). *Let the input be a valid finite identity-closure tagged instance with exact nonnegative mandatory weights, strictly positive optional weights, at least one carrier for every residual requirement, and a valid preprocessing reconstruction map. Complete enumeration returns a minimum-burden safe source sublibrary after checking all 2^n residual selections. If complete ties are requested, the map preserves all optimizer identities, and the output cap is not exceeded, it returns every original optimizer; otherwise it returns the declared stable representative.*

Proof. Every admissible library is a forced set plus one residual subset. Enumeration evaluates every such subset and takes the exact minimum; the fixed offset and valid lift preserve burden and feasibility. The final certificate maps tagged feasibility back to the original source semantics through Theorem 4.2. Complete reconstruction follows only under the stated map and output-cap conditions. Time is $2^n \text{poly}(|I|)$, so enumeration is exact but exponential in the number of optional strategies.

5.2.2 Requirement-mask dynamic programming

Encode a residual requirement subset by an r -bit mask. With $D_i(q)$ the least exact burden that attains mask q using the first i strategies, the recurrence is

$$D_i(q) = \min \left\{ D_{i-1}(q), \min_{p: \text{OR} R_i = q} (D_{i-1}(p) + w_i) \right\}. \quad (5.1)$$

Theorem 5.2 (Exact requirement-mask dynamic programming). *Let the input satisfy every assumption of Proposition 5.1. The recurrence in (5.1) reconstructs at least one minimum-burden safe source sublibrary. Representative mode uses $O(n2^r)$ recurrence operations, $O(2^r)$ exact cost states, and $O(n2^r)$ predecessor records when full reconstruction is retained. Value-only two-layer storage uses $O(2^r)$ memory. Materializing T tied optimizers adds $O(Tn)$ output bits and may be exponential in n . Its bit time is $2^r \text{poly}(|I|)$: the method is fixed-parameter tractable in r , but it is not polynomial in unrestricted input size.*

Proof sketch. Induct on i . At $i = 0$, only the empty subset attains mask zero. Every subset of the first i strategies either skips i , or removes i to give a predecessor mask and then takes it. The two transitions are exhaustive and construct valid subsets, so the recurrence stores the least exact burden for each mask. The all-ones mask is precisely residual feasibility. Offset, lifting, and Theorem 4.2 yield a safe global optimum. The state, transition, memory, and output counts give the displayed bounds. The complete induction and reconstruction proof is in Online Resource 1, Section S5.

5.3 Weighted greedy construction

After selecting every mandatory or forced strategy, let U^0 be the remaining tagged requirements and let

$$d := \max_i |R_i|, \quad H(d) := \sum_{j=1}^d \frac{1}{j}.$$

Theorem 5.3 (Transferred weighted-greedy guarantee). *Let the source be a finite identity-closure instance satisfying Theorem 4.2; let every residual requirement have a carrier; let mandatory weights be nonnegative and every optional weight be positive; and apply only mandatory/forced preprocessing with a valid exact offset and lift. If $U^0 \neq \emptyset$, weighted greedy returns a safe library of burden W_G satisfying*

$$W_G \leq H(d) \text{OPT} \leq H(r) \text{OPT}.$$

Reverse deletion returns a safe, inclusion-wise irreducible library with $W_{GR} \leq W_G$, so it satisfies the same bound. If $U^0 = \emptyset$, the forced library is optimal and the factor is one. The guarantee does not extend to arbitrary nonidentity closure.

Proof sketch. On the residual tagged universe, the rule minimizes weight per newly covered requirement. It is precisely the positive-cost weighted set-cover greedy algorithm

of Chvátal (1979), which gives $G \leq H(d) \text{OPT}_R$. Adding a nonnegative fixed burden W_0 preserves the displayed factor. Reverse deletion accepts only feasibility-preserving removals of positive-weight strategies and therefore cannot increase burden. Later removals cannot restore coverage lost by an earlier rejected removal, so one reverse pass is irreducible. The theorem transfers the classical guarantee through Theorem 4.2; it does not reprove the classical set-cover theorem. The cardinality variant replaces w_i by one in the score and has no retention-burden approximation guarantee when weights differ.

With straightforward incidence scans, greedy and reverse deletion each use a polynomial number of exact operations (at most n selection/deletion rounds and $O(nr)$ incidence work per round). This implementation bound does not alter the approximation factor.

5.4 Certified rechecked deletion

Rechecked deletion starts from the full source and never infers safety from a stale certificate. At every scan, each nonmandatory trial deletion is checked against the original source frontier and closure; after an accepted deletion, all candidate certificates are discarded and the scan restarts.

Deletion orders include heaviest-first, lightest-first, minimum unique-carrier exposure, source order, and a seeded random order. Maximum immediate additive release is the same rule as heaviest-first. The score chooses only among freshly certified safe deletions; the complete definitions are in Online Resource 1, Section S5.

Proposition 5.4 (Certified deletion endpoint). *Let the source be finite and safe relative to itself, let all mandatory strategies be protected, and suppose every accepted trial deletion is recomputed exactly and shown to preserve the original frontier, declared closure, and burden identity. Then every deterministic or seeded-random rechecked-deletion variant above terminates at a safe source sublibrary. A complete exact final scan certifies that no retained nonmandatory strategy can be safely deleted. This is a one-deletion irreducibility certificate, not a global-optimality or approximation certificate.*

Proof. The source is safe. Exact certification of each accepted trial preserves safety inductively, and each step removes one strategy, so the trace is finite. The final scan checks every remaining eligible deletion. The score and random order choose only among already certified candidates and therefore do not change the argument. If closure is nonidentity, this proposition requires the separate exact closure checker stipulated in its assumptions.

Multi-start deletion takes the lowest-burden certified endpoint over a fixed, registered number of random orders. Extra starts cannot worsen the best observed burden, but supply no approximation factor.

5.5 Worst-case local–global separation

Theorem 5.5 (Unbounded gap for heaviest-safe-first deletion). *Let $k \geq 2$ be an integer and let $\epsilon \in \mathbb{Q}$ satisfy $0 < \epsilon < k - 1$. Use one belief, identity closure, modules $m_1, \dots, m_k$, and a mandatory inactive strategy with zero profile, zero burden, and no modules. For each i , let singleton s_i have zero profile, weight one, and module set $\{m_i\}$. Let bundle s_B have zero profile, weight $1 + \epsilon$, and all k modules. Rechecked heaviest-safe-first deletes the uniquely heaviest safe bundle, terminates at the k -singleton library of burden k , and returns an inclusion-wise irreducible endpoint. The unique global optimum retains only the bundle and has burden $1 + \epsilon$. The exact ratio is*

$$\frac{k}{1 + \epsilon},$$

so heaviest-safe-first and one-deletion irreducibility alone have no constant burden guarantee over this family. The conclusion does not apply to every deletion order or every certified deletion heuristic.

Proof. All profiles are zero, so safety is exactly coverage of all modules. At the source, the bundle makes every singleton deletable and all singletons together make the bundle deletable. Since $\epsilon > 0$, the bundle is uniquely heaviest and is deleted first; afterward each singleton uniquely carries its module. Any safe library without the bundle must therefore retain all singletons and costs k , whereas the bundle alone is safe and costs $1 + \epsilon < k$; positive weights make it the unique optimum. Fixing, for example, $\epsilon = 1/2$ makes the ratio unbounded as k grows. A singleton-first order instead reaches the bundle optimum, proving the stated order boundary. The complete family proof and finite fixture are in Online Resource 1, Section S6.

5.6 Guarantee and evidence boundary

Enumeration and mask DP are exact finite methods under their stated identity-closure assumptions; the former is exponential in n , and the latter is exponential in r . Weighted greedy is approximate but has the transferred $H(d)$ bound; reverse deletion preserves that bound. Rechecked deletion and multi-start return exactly certified feasible, irreducible endpoints but remain heuristics. The MIP in Section 4.3 supplies solver evidence plus an exact returned-library postcheck. Complete proofs and machine-readable certificate schemas are in Online Resource 1; benchmark runtimes and method comparisons are experimental evidence reported separately in Section 7.

The negative guarantee concerns this named rule and one-deletion neighborhoods; it makes no claim about stronger exchange search.

6 Why current redundancy can destroy future value

The preservation constraint has an economic purpose beyond reproducing a static payoff menu. Frontier equality protects current operating opportunities; closure equality protects the inputs from which future opportunities can be generated. We separate current operating value from the value of access to a future project.

6.1 Operational and generative value

For a finite horizon, let $P_n(b, L)$ be the value when the library is frozen and research is unavailable, and let $U_n(b, L)$ be the corresponding full value when every feasible positive-duration project is available. Thus

$$\begin{aligned} P_0(b, L) &= 0, \\ P_{n+1}(b, L) &= F_L(b) + \beta \sum_{b'} P_B(b, b') P_n(b', L), \end{aligned} \tag{6.1}$$

where P_B is the filtered-belief transition under continued operation. Write $\Omega_n = U_n - P_n$ for the research-option premium. For inserting a candidate s , define total, operational, and generative increments by

$$\begin{aligned} \mathcal{I}_n(s \mid b, L) &:= U_n(b, L \cup \{s\}) - U_n(b, L), \\ \Delta_n^{\text{op}}(s \mid b, L) &:= P_n(b, L \cup \{s\}) - P_n(b, L), \\ \Delta_n^{\text{gen}}(s \mid b, L) &:= \Omega_n(b, L \cup \{s\}) - \Omega_n(b, L). \end{aligned} \tag{6.2}$$

Proposition 6.1 (Operational–generative decomposition). *For every finite horizon n , filtered belief b , finite raw library L , and admissible candidate s ,*

$$\mathcal{I}_n(s \mid b, L) = \Delta_n^{\text{op}}(s \mid b, L) + \Delta_n^{\text{gen}}(s \mid b, L). \tag{6.3}$$

If $F_{L \cup \{s\}} = F_L$, then the operational increment is zero. If in addition $C_{L \cup \{s\}} = C_L$, then the total increment is zero under the projection assumptions of Theorem 3.1.

The equality is an accounting identity, not a sign theorem. A strategy may be silent in the passive operating problem and valuable because it carries a module that enables research; conversely, the generative residual need not be positive if insertion changes costs or the relative appeal of projects. The frontier–closure projection makes the two channels operationally useful: frontier silence identifies zero current operating value, whereas joint frontier and closure silence identifies zero productive value within the declared process.

6.2 Sharp loss from frontier-only retention

Consider a dominated bridge strategy that uniquely carries the capability needed for one descendant project. Its deletion leaves the source frontier unchanged but makes the distinguished descendant unreachable. The following statement fixes the scope of the loss calculation: the incumbent operating block is common, there are no other differential projects or continuation gains, and the terminal continuation gain is bounded by C .

Theorem 6.2 (Sharp normalized frontier-only bridge loss). *Fix exact rationals $0 \leq \beta < 1$, $0 \leq \rho, \pi \leq 1$, a reward cap $C \geq 0$, a project cost $\kappa \geq 0$, and an integer delay*

$d \geq 1$. Suppose the bridge uniquely enables the descendant, retention gives survival mass ρ^d and conditional admission probability π , deletion makes that descendant mass zero, and $\kappa \leq \beta^d \rho^d \pi C$. For a realized continuation gain $g \in [0, C]$, with the option to decline research, the retained-minus-deleted optimal-value loss is

$$\text{Loss}_d(g) = \max\{0, \beta^d \rho^d \pi g - \kappa\} \leq \beta^d \rho^d \pi C - \kappa. \quad (6.4)$$

For fixed $(\beta, \rho, \pi, d, \kappa)$, the right-hand side is the sharp upper bound over descendant gains in $[0, C]$, attained at $g = C$. At that endpoint, whenever the denominator is positive, the loss divided by the attainable net bridge value equals one. If $C \leq 1$, the additive loss is at most one; arbitrarily large additive loss requires reward scaling.

The proof expands the one available project branch. Retention contributes the maximum of declining research and its discounted survival-and-admission mass times the continuation gain, less cost; deletion contributes zero. Maximization over the reward interval is attained at its upper endpoint. The factor $\rho^d \pi$ comes from sequential generation and conditional admission, not from an independence assumption. Common operating value cancels under frontier equality; if operation differs while research is in progress, its exact discounted difference must be added. The calculation above supplies the bridge proof; the positive-duration model accounts separately for any operating reward earned during research.

The result explains why operational redundancy is not generative redundancy. A frontier-only rule can be exact for today's operating menu and still remove the entire positive net value of a future bridge. It does not assert an unbounded normalized loss or a general value formula for arbitrary deleted libraries.

7 Registered computational study

The registered algorithmic compression benchmark is a finite synthetic study of tagged identity-closure instances. It can test exact-method agreement, certified burden quality, preprocessing prevalence, and the declared adversarial mechanisms. It cannot establish average-case complexity, steady-state runtime rankings, or a population law for financial libraries.

7.1 Registered design, execution, and audit

The final registry contains 173 analysis instances: 24 small exact, 128 structured, and 21 adversarial. Family B requested variation in strategy and tagged-row counts, coverage density, module overlap, bundle prevalence, unique-carrier frequency, weight dispersion, and frontier-module association. Family C contains the predeclared gap, tie, duplicate, dominance, rare-carrier, symmetry, and bundle mechanisms. Failed generation remains a terminal result; no failed row is silently repaired or dropped.

The algorithm grid contains enumeration, requirement-mask DP, two registered HiGHS formulations, weighted and cardinality greedy, greedy plus reverse deletion, rechecked deletion orders, seeded random deletion, and 32-start deletion. Maximum immediate additive burden release is exactly the heaviest-safe-first ordering. The two labels denote one method in all scientific comparisons.

The audit passed all 3907 registered run units and accepted 3526 returned libraries after original-coordinate mandatory, coverage, frontier, identity-closure, and exact-burden checks. It retained 381 unsuccessful records in their terminal categories. HiGHS OPTIMAL remains solver evidence plus an exact incumbent check, not formal or exhaustive proof.

7.2 Exact agreement and applicability

Among 38 registry instances, all required exact-method burdens were available and equal on 31; 7 did not have a complete comparison. 6 adversarial enumeration jobs are recorded as `IMPLEMENTATION_ERROR`, but their substantive cause is known: the residual optional count exceeded the declared enumeration cap. The recorded status is retained in the failure census. They are enumeration-inapplicable under that cap, and exact agreement is unavailable rather than inferred. Optimizer identities differed at equal burden on 9 complete comparisons, which records valid tied optima rather than disagreement.

These are exact finite facts about specified instances. The medium structured references are HiGHS-optimal incumbents that pass exact feasibility and burden rechecks; no second solver was pinned.

7.3 Certified burden quality

Table 2 separates Family A exact finite references from Family B exactly rechecked HiGHS-optimal references. Within those distinct evidence classes, weighted greedy plus reverse deletion and heaviest-safe-first often attain the available reference, while several other deletion orders have larger observed tails. Multi-start attains every available Family A reference but not every Family B reference. These are descriptive outcomes of the registered instances, not approximation guarantees or an algorithm ranking beyond the design.

The registered heaviest-safe-first gap family gives a different kind of evidence. Its exactly rechecked finite ratios match Theorem 5.5 at every registered scale. The theorem establishes the universal family statement; the experiment checks the implementation on the named fixtures.

The supporting benchmark appendix reports the finite ratios.

7.4 Preprocessing and realized design limits

Full fixed-point preprocessing has a strong realized effect. In every structured size cell, median variable and requirement reduction is complete among available records. This is evidence that forced-carrier propagation is important for these generators. It is simultaneously a limitation: most structured instances do not exercise a difficult residual MIP or DP search, so the study cannot identify general residual-search scaling.

The registered generator also contains infeasible requested cells. All 352 run units arising from 16 structured source-instance cells are retained, and the artifact audit shows that the failures remove complete combinations of low density and bundle/unique-carrier settings rather than occurring at random. Effects and interactions requiring those absent cells are not estimable from this design.

Algorithm	Attained/reference	Mean gap	Median	q90	Worst
<i>Family A: exact finite reference</i>					
Weighted greedy	16/23	0.08684	0	0.25	0.5
Cardinality greedy	13/23	0.6375	0	1.4	7
Weighted greedy + reverse deletion	20/23	0.02923	0	0.16	0.25
Heaviest safe first	22/23	0.008696	0	0	0.2
Lightest safe first	5/23	1.586	1	3.32	7
Maximum immediate release	22/23	0.008696	0	0	0.2
Minimum unique-carrier exposure	17/23	0.2303	0	0.2686	3
Declared source order	11/23	0.6965	0.2857	2	3
Random-order deletion	8/23	1.384	0.6667	3.3	7
32-start random deletion	23/23	0	0	0	0
<i>Family B: HiGHS-OPTIMAL, exactly rechecked reference</i>					
Weighted greedy	82/111	0.02982	0	0.06452	0.5
Cardinality greedy	24/111	0.8904	0.25	2	7.5
Weighted greedy + reverse deletion	103/111	0.01038	0	0	0.5
Heaviest safe first	94/111	0.09983	0	0.2	3
Lightest safe first	11/111	4.128	0.7143	15	63
Maximum immediate release	94/111	0.09983	0	0.2	3
Minimum unique-carrier exposure	98/111	0.04435	0	0.05556	1
Declared source order	36/111	1.533	0.1818	5	20
Random-order deletion	28/111	1.753	0.2222	6.8	17
32-start random deletion	76/111	0.1324	0	0.5	2.667

Table 2 Registered heuristic burden quality. Relative gap is $(W - W^*)/W^*$. Attainment denominators include every registered row with an available reference; gap moments use exactly feasible returned libraries. Family A and Family B have different reference evidence and are deliberately not pooled.

Size	Preprocess n	Median var. red.	Median row red.	Median log PAR-2 ratio	32-start complete	Improved
N64-R32	28/32	1	1	1.187	28/32	20/28
N64-R64	28/32	1	1	1.29	28/32	19/28
N192-R32	28/32	1	1	2.031	15/32	12/15
N192-R64	28/32	1	1	2.243	27/32	21/27

Table 3 Registered preprocessing and multi-start evidence on structured instances. Reductions are shares removed. The log ratio is $\log(\text{PAR2}_{\text{full}}/\text{PAR2}_{\text{mandatory}})$, so a positive value favors the mandatory-only pipeline in elapsed penalized runtime despite the model-size reduction. Unavailable multi-start endpoints are not coded as zero improvement.

Multi-start deletion completes all registered starts on 143 instances and improves on the first start on 96. Memory-limited primary runs remain failures; successful sensitivity records are not substituted into their denominators.

7.5 Limits of the comparison

Recorded times include first-call Julia compilation and contention among eight one-thread workers on the recorded worktree. They do not support steady-state speed rankings. The registered performance profile also mixes feasible-return and completed-search targets, so it is not comparative evidence here. Registered model outputs do not support structural inference: their HC3 uncertainty ignores repeated observations

within instances, and entire requested generator cells are unavailable. Neither model refitting nor missing outcomes are used to improve the reported comparison.

Every unsuccessful run remains in the full denominator, including generation failures, memory limits, interruptions, absent primals, and nonbinary returned vectors. No run ended at the time limit. Online Resource 1, Section S7 gives the design, complete aggregate failure census, and exact-agreement audit.

8 Financial evidence: point-in-time portfolio-library retention

The financial study asks whether innovation-safe compression can reduce the active validation burden of portfolio-strategy libraries while preserving a larger menu of admissible future research options than a budget-matched frontier-only library. It is a retrospective development-facing panel, not an independent confirmatory market test. Its contribution is the combination of point-in-time source construction, an explicit portfolio grammar, prespecified proposal decisions, exact structural rechecks, and a transparent held-out null result.

8.1 Sequential information boundary

At each annual origin y , early historical blocks construct the source portfolio-strategy library and its operating frontier; the subsequent predecision block estimates compression profiles and solves the safe and frontier-only arms; year $y + 1$ enumerates closure-enabled proposal options and freezes both policy choices; and year $y + 2$ supplies the held-out portfolio return paths. Proposal outcomes do not enter compression. Evaluation outcomes do not enter compression, option generation, uncertainty estimation, adoption, or any policy choice. The full candidate ledger retains unavailable and rejected candidates rather than conditioning the analysis on successful selection.

The primary universe is formed from the most liquid eligible US common equities at each compression origin. A separate plain-ETF universe is the registered replication when its point-in-time eligibility gate passes. Eligibility never requires survival into the proposal or evaluation window; observed delisting returns are retained and a terminated position subsequently moves to cash. The single failed ETF universe gate remains in the registered denominator. Raw security identifiers, prices, and returns remain inside the licensed boundary.

Each source library applies a fixed factorial grammar of portfolio-level signals, filters, holding horizons, allocation rules, gross-exposure targets, and exits to the same contemporaneous universe. Closure consists of 31 declared rule, interface, data, corporate-action, portfolio-aggregation, and cost/capacity capabilities. These declarations are more operationally concrete than free-form module tags, but they remain a modeled grammar: they do not prove that tacit knowledge or every institutional artifact is frictionlessly recoverable.

8.2 Compression and proposal screening policy

The innovation-safe arm minimizes the registered validation-computation burden subject to exact source-frontier and full declared-closure equality. Its frontier-only comparator is augmented under the safe arm’s realized burden cap using a frozen preproposal order. The comparator action set is a subset of the safe action set in every gate-passed cell. Cash is always feasible, and the safe policy begins with the exact comparator choice. It can replace that choice only when the registered simultaneous moving-block bootstrap score for incremental proposal value exceeds 0.0025 in annual certainty-equivalent return. The policy therefore treats closure as an option, not a forced trade. The score uses the reflected resampling statistic documented in Online Resource 1, Section S8. Its nominal 90% tuning level has no established finite-sample simultaneous coverage for this design. We interpret it as a prespecified adoption screen, not a verified confidence guarantee or protection against an adverse held-out realization.

Across the gate-passed origins, 3 safe solutions are resolved by exact preprocessing and 34 are returned with HiGHS OPTIMAL status. Every result is independently reconstructed and rechecked for mandatory inactive retention, exact table frontier equality, exact identity-closure equality, and exact integer burden. Those rechecks certify feasibility and burden; for the solver-resolved cells, global optimality remains mixed-integer solver evidence rather than a formal or exhaustive proof.

Table 4 gives the aggregate structural and held-out results. Innovation-safe retention reduces the mean source burden index by 73.1% in common equities and 74.7% in ETFs while preserving all registered capabilities. Relative to the budget-matched frontier-only library, it exposes on average 65.95 additional common-equity proposal actions and 66.06 additional ETF actions.

Figure 2 separates burden reduction and option availability from the origin-level held-out outcomes.

8.3 Held-out result: options are not exercises

The screening policy makes 0 common-equity replacements. Consequently, every common-equity held-out innovation-safe-minus-frontier-only contrast is exactly zero because the two policies choose the same action. This is not evidence that the two libraries are economically equivalent; it is evidence that none of the additional options clears the proposal-time uncertainty rule.

Universe	Origins	Mean source burden	Mean safe burden	Mean reduction	Added options	Held-out CE difference
Common equity	19	8169.2	2183.8	73.1%	65.95	0.0000
Plain ETF	18	8909.2	2231.7	74.7%	66.06	-0.0022

Table 4 Point-in-time portfolio-library retention results. Burden is the registered validation-computation index, not currency. Added options are safe-library proposal actions absent from the budget-matched frontier-only library. The held-out column is innovation-safe minus frontier-only certainty-equivalent difference: all 19 common-equity choices coincide, while one of 18 complete ETF choices differs. Safe solutions pass exact original-instance frontier, closure, inactive-retention, and burden rechecks; solver-reported optimality remains solver evidence.

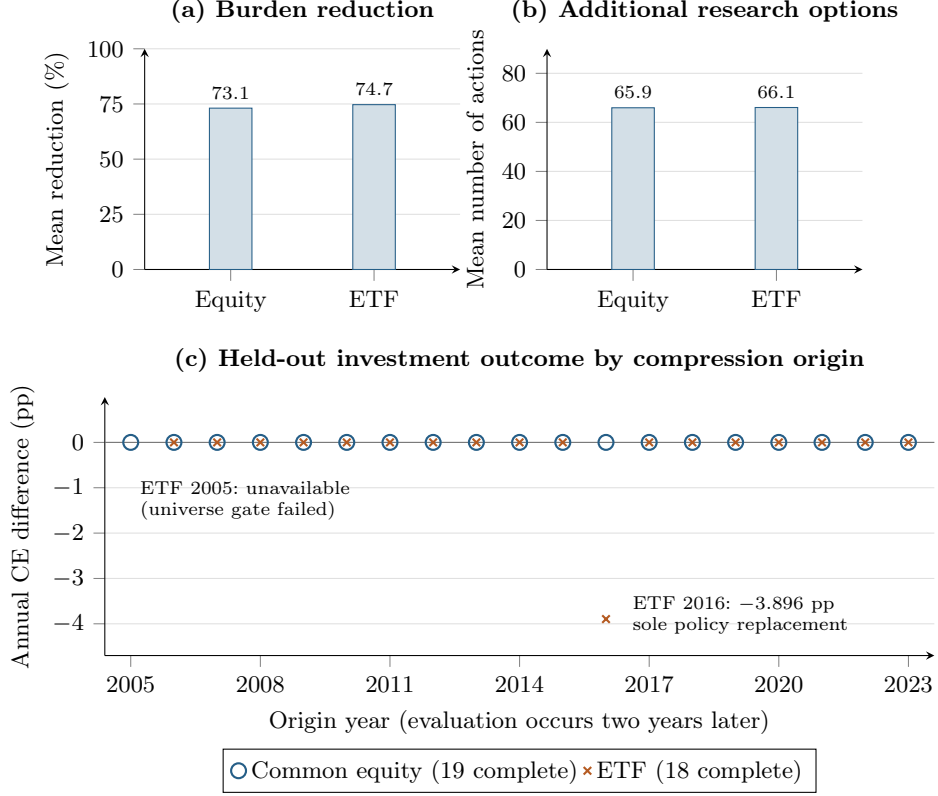


Fig. 2 Structural savings and options are separate from investment gains. Panels (a)–(b) average across 19 common-equity and 18 complete ETF origins. Burden reduction is relative to each origin’s source; added options are relative to its budget-matched frontier-only library. Burden is a validation-computation index, not monetary cost. Panel (c) reports every complete origin’s safe-minus- frontier-only annual certainty-equivalent (CE) difference in percentage points. All common-equity choices and 17 ETF choices coincide, so their contrasts are exactly zero; the circles and crosses overlap at zero. The failed ETF origin is missing, not zero. These are descriptive outcomes, without a cross-origin confidence interval. Source: sealed public predecision and evaluation manifests.

The ETF replication makes 1 replacement. That replacement realizes a held-out certainty-equivalent difference of -0.0390 , producing a mean of -0.0022 over the complete ETF origins. All other complete ETF pairs choose the same action. The negative observation is retained and is not offset by reporting a favorable post hoc selector. Full origin-level results, the prespecified robustness grid, negative-control policies, search-bias diagnostics, and capacity records appear in Online Resource 1.

The contrast is a postdecision portfolio certainty-equivalent diagnostic under the registered risk and cost rule. It is not an alpha estimate, a forecast, or a deployable-performance claim. In particular, the many structural options do not count as independent financial observations, and the safe policy’s abstention cannot be reinterpreted as positive performance.

8.4 Robustness, licensed-data boundary, and managerial implication

The proposal robustness analysis retains the full prespecified grid, including unavailable cells. Common-equity safe and comparator choices remain identical throughout that grid. ETF differences remain confined to the same origin and disappear at higher transaction-cost settings. A closure-sham analysis that was not frozen before proposal outcomes became known is explicitly unavailable, not reconstructed post hoc. Capacity analysis scores frozen choices without reselection and retains missing-source, incomplete-path, and hard participation- cap failures in its denominators.

Licensed daily panels were used locally to construct the point-in-time universes and portfolio paths. The public artifact redistributes only aggregate structural manifests, cryptographic seals, permitted summaries, and synthetic records. It contains no raw CRSP/WRDS rows, security identifiers, or return paths. Independent replication of the licensed stages therefore requires separate data access, whereas all published table values can be regenerated from the committed public-safe manifests.

For research governance, the result is useful precisely because it is not a success story about returns. A manager can remove substantial declared validation burden while exactly preserving current opportunities and a much larger research-option menu. Whether any added option should be exercised is a separate statistical decision, and the registered selector can abstain. Removal from the active maintenance set also does not authorize deletion from legal, compliance, or reproducibility archives. A stronger empirical claim would require a newly locked market vintage or independent panel, documented real capability inventories, and monetary maintenance costs calibrated without using postdecision outcomes.

8.5 Calibrated closure-option mechanism experiment

The theorem is universal under its assumptions, but it does not say whether a finite-data policy can recognize a valuable bridge. We therefore isolate that question in a design-locked synthetic experiment. The innovation-safe and frontier-only libraries have identical current frontiers and identical retention burdens. Only the safe library carries the capability needed to run one declared research project. For delay d , survival probability ρ , admission probability π , descendant gain G , and research cost κ , the registered treatment margin is

$$M = \beta^d \rho^d \pi G - \kappa.$$

The safe oracle researches exactly when $M > 0$; the frontier-only project is infeasible. A learner observes an independent noisy proposal signal and adopts only when it exceeds a threshold calibrated under the null.

The noise law uses only aggregate nuisance estimates from 2479 complete proposal paths and 623647 within-cell residual observations in the point-in-time financial study. Cross-strategy date means are removed before estimating volatility, lag-one dependence, and tail thickness. These quantities calibrate noise and dependence, not the sign of any regime. The test-world seeds are disjoint from calibration, and the powered margin is

fixed at twice the null threshold before test simulation. Thus its size is a power-design dose, not an estimated return premium.

Table 5 reports all four prespecified regimes. At the low positive dose, the oracle value is positive but the learned-value interval crosses zero and adoption is only 7.35%. At the powered dose $M = 0.0540$, the learned mean is 0.0512 and adoption is 94.17%. The null point adoption rate, 5.76%, satisfies the registered 6.5% finite-test gate; its Wilson upper endpoint is 6.52%, so the interval is not wholly below that threshold. Every world passes the equal-frontier, equal-burden, and bridge-availability structural checks.

Figure 3 shows the oracle–learner gap with uncertainty and a separate view of the low-value regimes.

This evidence makes the option/exercise distinction concrete. Exact closure preservation can create access to a genuinely positive project while a threshold-based learner rarely exercises a small option at the available signal-to-noise ratio. Conversely, the powered row shows recovery only for a deliberately larger treatment. The experiment validates the declared bridge mechanism under calibrated synthetic noise. It is not a historical market test, a causal return estimate, an alpha estimate, or a claim that the powered margin is empirically prevalent.

9 Relation to existing work

The financial motivation connects costly information acquisition with the maintenance of productive capabilities. [Van Nieuwerburgh and Veldkamp \(2010\)](#) jointly model information acquisition and portfolio choice, showing how learning incentives affect investment. [Eisfeldt and Papanikolaou \(2013\)](#) model organization capital embodied in key talent and link its firm-specific productivity and outside options to financial risk. Our retention problem focuses on declared reusable inputs to later strategy research: removing a carrier can change which projects remain feasible even when its current operating contribution is redundant. The closure is a specified capability inventory, not an estimate of tacit knowledge or an institutional measure of organization capital. This paper asks when a smaller library preserves that inventory and its modeled uses; it does not determine whether every capability is worth maintaining.

Regime	True margin	Oracle mean [95% CI]	Learned mean [95% CI]	Adoption
Adverse margin	-0.0270	0.0000 [0.0000, 0.0000]	-0.0000 [-0.0000, 0.0000]	0.05%
Null margin	0.0000	0.0000 [0.0000, 0.0000]	-0.0002 [-0.0003, 0.0000]	5.76%
Low positive dose	0.0025	0.0026 [0.0019, 0.0032]	0.0001 [-0.0001, 0.0002]	7.35%
Powered positive dose	0.0540	0.0541 [0.0529, 0.0554]	0.0512 [0.0500, 0.0525]	94.17%

Table 5 Calibrated closure-option mechanism experiment, with 4096 synthetic worlds per prespecified regime. Current frontier and burden are equal by construction; only the innovation-safe library can admit the designated bridge descendant. Intervals are registered finite-simulation uncertainty summaries. This is synthetic mechanism evidence, not a market-return, alpha, forecasting, or causal estimate.

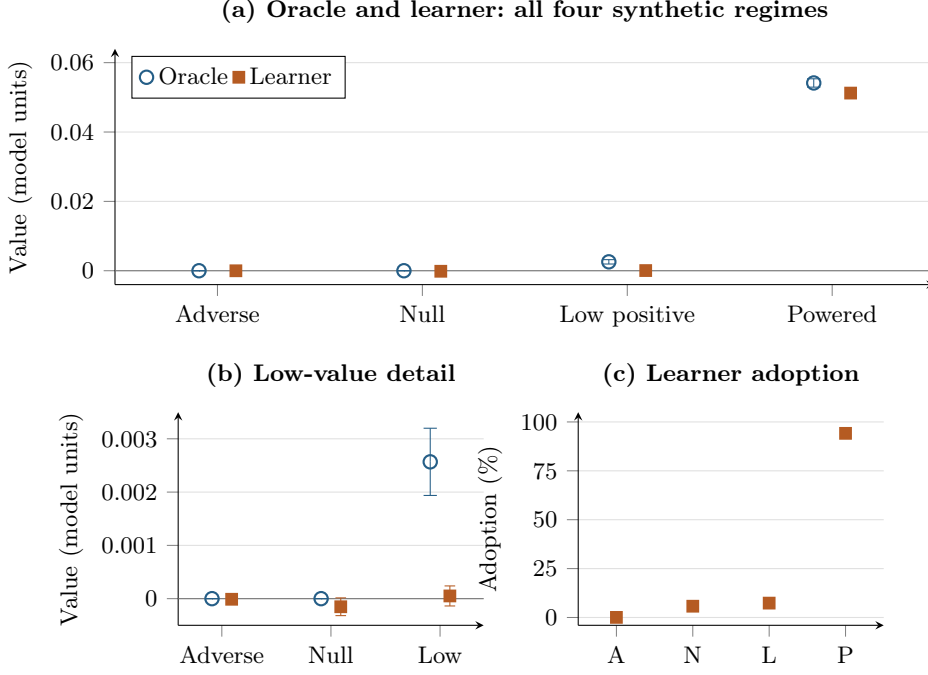


Fig. 3 Synthetic option value and exercise under calibrated noise. Each regime contains 4,096 test worlds. Panel (a) plots value means with 95% intervals (mean ± 1.96 standard errors); panel (b) repeats the first three regimes on an explicitly expanded value scale. Panel (c) shows adoption with 95% Wilson intervals, in the same adverse (A), null (N), low (L), and powered (P) order. Zero-length oracle intervals at the adverse and null doses reflect deterministic abstention. The true margins are -0.027008 , 0 , 0.0025 , and 0.054016 . Common random numbers pair regimes. The powered dose is chosen by the power design, not estimated from market returns. All plotted values and intervals are regenerated from the complete sealed synthetic ledger.

Belief-state reduction supplies the control architecture used here. [Smallwood and Sondik \(1973\)](#) and [Sondik \(1978\)](#) develop finite- and infinite-horizon partially observed control; [White \(1976\)](#) treats a partially observed semi-Markov process. Our additional question concerns the library component of state: which retained strategies leave both operating choices and the declared generation process unchanged? The frontier-closure pair summarizes that component under the factorization assumption. The belief and active-project variables remain in the controlled state.

The preservation target distinguishes this problem from nearby reductions. Incremental pruning removes redundant value vectors in a fixed POMDP backup ([Cassandra et al. 1997](#)); state aggregation identifies behaviorally equivalent states for a specified model ([Givan et al. 2003](#)). Neither requires a removed strategy to remain available as a carrier for subsequent generation. Compact control libraries trade static usefulness against library size and search effort ([Dey et al. 2012](#)). Reward-free policy compression uses set cover to represent state-action distributions for downstream tasks ([Mutti et al. 2022](#)). Here the choices are subsets of a given source, and the preserved objects are its operating frontier and generative closure.

Once identity closure makes those objects a finite tagged requirement set, the algorithmic core is classical weighted cover. The complexity reduction uses Karp’s set-cover problem (Karp 1972), and the greedy guarantee is transferred from Chvátal (1979). The modeling contribution is the mapping from the financial preservation condition to that problem; the generic cover algorithms are established methods. Nonidentity closure requires a separate representation of capability interactions. Maximum-coverage or adaptive selection guarantees do not follow merely from calling capabilities a library.

In finance, automated construction has long used genetic search and reinforcement methods (Allen and Karjalainen 1999; Moody and Saffell 2001). Those studies concern constructing or operating strategies; our decision is which entries in a maintained source can be removed while preserving its specified opportunities. Our burden index measures the declared maintenance requirements; it is not an estimate of institutional compliance costs.

A larger candidate menu also creates a larger selection problem. The literature on data snooping and multiple testing explains why more available strategies cannot itself establish better investment performance (Sullivan et al. 1999; White 2000; Harvey et al. 2016). The financial study therefore freezes a bootstrap-based proposal-time adoption rule before held-out evaluation and reports abstention and adverse exercise. Its known historical setting still limits external validity, and post-publication decay provides a further reason for restraint (McLean and Pontiff 2016).

10 Conclusion

A strategy can be redundant for current operation while remaining useful as a carrier of a future research capability. A manager who wants to reduce active maintenance burden must specify both roles before deleting entries. Under the declared finite factorization, preserving the operating frontier and generative closure preserves the library’s productive process. The theorem’s assumptions are substantive: dependence on a named strategy or another omitted attribute can invalidate that reduction.

Under identity closure, minimum-burden safe retention is a tagged weighted cover. This gives access to established exact and approximate methods, with independent checks in the original strategy coordinates. Safe local deletion alone provides no general burden guarantee, as the heaviest-first family shows. The registered benchmark confirms exact-method agreement where available and documents burden quality and preprocessing effects within its realized design; it does not support general speed rankings.

The point-in-time application preserves many research options at lower declared burden, yet the common-equity screening policy makes no additional investment and its one ETF replacement loses held-out value. The synthetic experiment shows that small positive bridge value can also be difficult to detect under the specified noise. Preserving an option therefore leaves a separate statistical and economic decision about exercise.

The natural extensions are capability interactions beyond identity closure and an independent financial study with documented institutional capabilities and maintenance costs measured before future outcomes. The present findings support a precise retention contract, rather than a claim of alpha or deployable performance.

Supplementary information

Online Resource 1, *Supporting Proofs, Methods, and Reproduction*, contains supporting proofs and counterexamples, complete benchmark outcome accounting, financial and synthetic methods, origin-level results, and a reproduction guide. The combined preprint appends it with S-prefixed numbering. The editable package includes the document inputs and the public evidence used here.

Statements and Declarations

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Ethics and consent. Not applicable; the study does not involve human participants or animals.

Data availability. The accompanying source package includes exact tables, public aggregate financial outputs, benchmark records, and the synthetic experiment ledger. Raw and row-level CRSP/WRDS observations are excluded; reproducing the licensed stages requires independent licensed access. The reproduction guide identifies the required inputs and source files.

Code availability. Julia 1.12.6 and Lean 4.32.0 sources and pinned environments are available at <https://github.com/dramirez37/financial-strategy-library-compression>. The [versioned preprint package](#) contains the PDFs, editable source, and public evidence. Its reproduction manifest binds the selected scientific sources by hash. The document archive builds the PDFs independently; full computational replication uses the specified repository sources. Original software is MIT licensed and manuscript material is CC BY 4.0, subject to the licensed-data and third-party exclusions in **LICENSE**.

Author contributions. David Ramirez conceived the study, developed the theory and software, designed and conducted the analyses, curated the public artifacts, and prepared the manuscript.

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